

GROW: Watermark Generation with Progressive Guidance for Diffusion Models

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Abstract

Digital watermarking is a cornerstone for copyright protection. With the rapid advancement of generative models like diffusion models, in-generation and training-free watermarking techniques have garnered more attention for their endogeneity and convenience. These methods typically embed a watermark into the initial noise, where watermark extraction relies on Denoising Diffusion Implicit Models (DDIM) inversion. However, the computationally intensive extraction process severely hinders their path toward practical deployment. To overcome this critical bottleneck, we propose GROW, a novel training-free paradigm that reframes watermarking from a one-shot “embedding” to a progressive “growth”. By progressively guiding using frequency-domain gradients, GROW naturally weaves the watermark into the image, which enables inversion-free extraction. Comprehensive experiments on multiple datasets show that GROW not only achieves superior robustness and imperceptibility but also offers a detection speed nearly 100x faster than inversion-based techniques. The code will be made publicly available.

1. Introduction

The rapid advancement of generative models, particularly diffusion models (DMs) [19, 29, 33, 36, 37] has revolutionized AI-generated content (AIGC), enabling the creation of photorealistic images from textual prompts. This accessibility, however, brings significant challenges in copyright protection and content traceability [5, 22]. Digital watermarking [20] is a critical technology to address these issues by embedding imperceptible information into generated images, thereby providing a means for authentication and provenance tracking.

Existing watermarking methods for diffusion models can be broadly classified into three categories. *Post-processing*

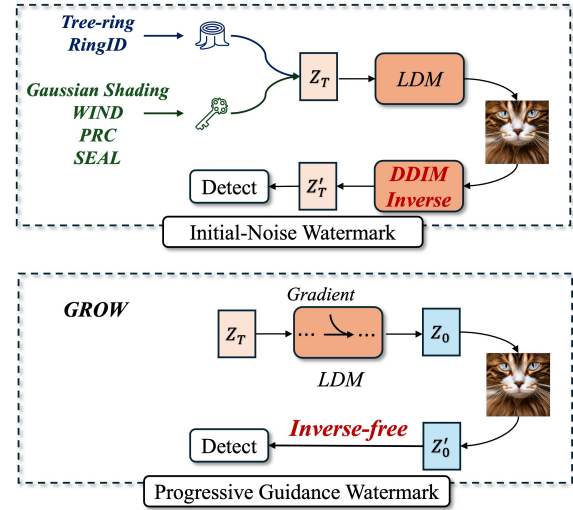


Figure 1. A comparison of watermarking paradigms for diffusion models. **(a) Initial-Noise Watermarking:** The dominant paradigm, based on passive scattering, embeds a watermark in the initial noise and relies on costly inversion for extraction. **(b) Progressive Guidance Watermarking:** Our new paradigm GROW, based on active guidance, progressively guides the generation process, which enables inversion-free extraction.

methods [6, 7, 35, 41, 44] directly alter the final image, often at the cost of visual quality and robustness. In contrast, in-generation watermarking integrates the watermark directly into the generation process. This provides a crucial advantage of endogeneity: the watermark is semantically hidden within the image. Consequently, its complete removal would require knowledge of the diffusion model’s internal parameters, access to which is typically impractical. Within in-generation methods, *Fine-tuning-based* approaches [4, 8, 10, 11, 13, 14, 24, 27, 28, 31, 39, 43] embed the watermark by either modifying the model’s parameters or introducing it as a conditional input, which requires extensive computational resources for retraining. More recently, *Training-free* methods [2, 3, 9, 15, 21, 23, 25, 34,

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[38, 40] have emerged as a flexible alternative. These methods typically operate by manipulating the initial noise latent before the denoising process begins.

A seminal work in this area, Tree-Ring [38], embeds watermarks by creating specific patterns in the initial noise latent. This foundational idea has inspired a series of subsequent methods [2, 3, 9, 15, 21, 23, 25, 34, 40] that also operate on the initial noise. While ingenious, these initial-noise-based methods share a critical flaw: their reliance on a computationally expensive DDIM inversion process [36] to recover the initial noise for watermark detection. This severe latency bottleneck hinders their practicality for large-scale, real-time applications. Instead of embedding watermark into initial noise, a naive approach might be to add the watermark to the final latent representation just before decoding. However, our experiments in Figure 6 show that this drastically disrupts the learned data distribution, leading to severe image degradation.

We realize that the solution lies not in abandoning the principle of endogeneity, but in rethinking its mechanism. The robustness of initial-noise-based methods stems from a **passive scattering** process, where the denoising model implicitly filters a watermarked noise, fusing a portion of it with the image semantics. Our key insight is to achieve this same desirable fusion through **active guidance**. As depicted in Figure 1(b), we bridge image generation and watermark generation by leveraging the model’s own iterative denoising process. Instead of modifying the initial noise, we introduce a gradient-based guidance mechanism. At each denoising step, we compute the difference between the latent features (in the frequency domain) and a fixed target watermark pattern. The gradient of this difference is then used to gently steer the generation, constituting a progressive guidance approach. The subtle perturbations introduced at each step accumulate along the model’s learned data manifold. In subsequent steps, the U-Net denoiser, driven by the conditioning signal, naturally incorporates these perturbations into the emerging image textures like feathers and tree bark, resulting in a deeply fused watermark.

This reformulation forms the core of our method, **GROW** (Generation with **pro**gressive **W**atermarking). Crucially, it retains the semantic fusion advantage of prior work while offering a decisive practical benefit: extraction is direct, efficient, and completely free from inversion. GROW thus establishes a new paradigm, shifting from a static “imprint” to a dynamic “co-evolution” between the image and the watermark.

Our work makes three key contributions:

- We propose GROW, a novel training-free paradigm that reframes watermarking as a progressive guidance process, eliminating the dependency on costly model inversion for extraction.

- We develop a novel mechanism that leverages gradient guidance in the frequency domain to weave the watermark into the final latent representations.
- We empirically demonstrate that GROW not only surpasses existing methods in robustness and imperceptibility but also achieves a nearly 100x speedup in extraction, offers a fast and practical watermarking solution.

2. Related Works

Recent research on image watermarking can be broadly categorized into three categories.

Post-processing-based: These watermarking algorithms [6, 7, 35, 41, 44] are applied directly to the final generated images. These methods, often adapted from classical techniques [35, 44], typically operate in the spatial domain, a transform domain (e.g., DCT, DWT) or a latent space to embed the watermark signal. For example, RivaGAN [41] uses a GAN-based framework with an encoder-decoder architecture to embed and extract watermarks.

Fine-tuning-based: To achieve deeper integration, approaches amalgamate the watermarking process with the model’s architecture [4, 8, 10, 11, 13, 14, 24, 27, 28, 31, 39, 43]. These approaches typically modify a pre-trained diffusion model to instill a watermarking capability. For instance, Stable Signature [14] fine-tunes the VAE decoder with a dedicated watermark extractor, enabling the decoder to implicitly embed a signature during image reconstruction. Other works [27, 43] propose fine-tuning a specific trigger to hidden watermark. Some methods also introduce the watermark as a conditional input during the training phase [39]. Although these methods can produce high-quality watermarked images, they require significant computational resources for retraining and result in modified model parameters, which can be a major barrier for practical deployment.

Training-free: Recently, training-free methods [2, 3, 9, 15, 21, 23, 25, 34, 40] have emerged as a flexible, plug-and-play alternative. Their dominant strategy, as illustrated in Figure 1(a), is to manipulate the initial noise latent $\mathbf{z}_T$. The seminal work, Tree-Ring [38], demonstrated that watermark information can be embedded in the initial noise. This foundational idea has inspired a lineage of research. For instance, WIND [2] focuses on designing specific patterns for improved robustness, while SEAL [3] innovatively links the watermark to the semantic content of the input prompt. Gaussian Shading [40] proposes mapping the watermark to a distribution-preserving Gaussian representation to achieve performance-lossless embedding. Despite their ingenuity, a critical flaw shared by all these initial-noise-based methods is their reliance on a computationally expensive inversion process [36] for watermark detection. Recovering the initial noise latent from a given image is a time-consuming procedure that severely hinders the prac-

ticality of these methods for large-scale, real-time applications. Our work GROW, belongs to the training-free category but introduces a fundamentally different paradigm that circumvents this inversion requirement entirely. We present a theoretical analysis in supplementary material.

3. Preliminaries

3.1. Latent Diffusion Models

Latent Diffusion Models (LDMs) [33] generate images by iteratively denoising a random latent variable $\mathbf{z}_T$ using a U-Net model. A key step in modern schedulers is predicting the "clean" latent $\hat{\mathbf{z}}_0$ from the noisy latent $\mathbf{z}_t$ and the predicted noise $\mathcal{E}_\theta(\mathbf{z}_t, t, c)$:

$$\hat{\mathbf{z}}_0 = \frac{1}{\sqrt{\bar{\alpha}_t}}(\mathbf{z}_t - \sqrt{1 - \bar{\alpha}_t}\mathcal{E}_\theta(\mathbf{z}_t, t, c)), \quad (1)$$

where $\bar{\alpha}_t$ is a noise schedule parameter and c is the conditioning. Our method leverages this predicted $\hat{\mathbf{z}}_0$ as the target for guidance. The final latent $\mathbf{z}_0$ is then decoded into an image $\mathbf{x} = \mathcal{D}(\mathbf{z}_0)$ by a VAE decoder $\mathcal{D}$.

3.2. Guidance in Diffusion Models

Guidance techniques are fundamental to controlling the generation process in diffusion models.

Classifier Guidance [12], an early approach, steers generation by incorporating gradients from a pre-trained classifier $p_\phi(y|\mathbf{z}_t)$ into the noise prediction:

$$\hat{\mathcal{E}} = \mathcal{E}_\theta(\mathbf{z}_t, t) - \sigma_t \nabla_{\mathbf{z}_t} \log p_\phi(y|\mathbf{z}_t) \quad (2)$$

where $\mathcal{E}_\theta(\mathbf{z}_t, t)$ is the model's predicted noise, σ_t is the noise schedule, and $\nabla_{\mathbf{z}_t} \log p_\phi(y|\mathbf{z}_t)$ is the gradient of the classifier with respect to the latent $\mathbf{z}_t$ for class y . While powerful, this method requires training a separate classifier on noisy data and can compromise sample diversity.

Classifier-Free Guidance (CFG) [18] has become the mainstream approach by circumventing the need for a separate classifier. It achieves this by jointly training the diffusion model on conditional and unconditional inputs and interpolating their noise predictions during inference:

$$\hat{\mathcal{E}}_{\text{CFG}} = \mathcal{E}_\theta(\mathbf{z}_t, t, \emptyset) + w \cdot (\mathcal{E}_\theta(\mathbf{z}_t, t, c) - \mathcal{E}_\theta(\mathbf{z}_t, t, \emptyset)) \quad (3)$$

Here, $\mathcal{E}_\theta(\mathbf{z}_t, t, c)$ is the conditional prediction (with prompt c), $\mathcal{E}_\theta(\mathbf{z}_t, t, \emptyset)$ is the unconditional prediction (with empty prompt $\emptyset$), and $w > 1$ is the guidance scale that amplifies the direction pointed by the conditional prediction. This allows for flexible control over the trade-off between sample quality and adherence to the condition.

Our work is conceptually inspired by these guidance paradigms. We adapt the core idea of steering the denoising trajectory to the novel task of invisible watermarking. Instead of guiding semantic attributes like "an astronaut riding a horse," we guide the generation process to subtly and

progressively incorporate a watermark signal, leveraging a similar mechanism to achieve deep fusion within the natural image manifold.

4. Method

4.1. Overview

Our proposed method, GROW, introduces a new paradigm for training-free watermarking. Instead of embedding a static pattern into the initial noise, we progressively guide the generation process towards a predefined watermark target. This is achieved by treating watermarking as an optimization problem solved concurrently with denoising. The overall framework is depicted in Figure 2.

4.2. GROW

4.2.1. Frequency-Domain Watermark Encoding

To create a watermark that is both imperceptible and robust, we embed it in the frequency domain of the latent space using the Discrete Cosine Transform (DCT). We strategically choose to embed the watermark in the mid-frequency band. This choice is crucial: embedding in low-frequency components would visibly distort the core structure of the image, while embedding in high-frequency components would make the watermark vulnerable to removal by common operations like JPEG compression or blurring. The mid-frequency band thus offers an optimal trade-off between visual quality and robustness.

First, a secret message W is converted into a binary bit sequence w . Given a secret key K , we use a seeded pseudo-random number generator to generate a binary mask matrix $\mathbf{M}$. The bit sequence w is repeated and mapped to mask matrix to define a target signal matrix $\mathbf{S}$:

$$\mathbf{S}(u, v) = \begin{cases} \alpha \cdot (2w_i - 1) & \text{if } \mathbf{M}(u, v) = 1 \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

where w_i is the bit assigned to the coordinate (u, v) , and α is the watermark strength.

4.2.2. Progressive Watermark Guidance

This is the core of our method. Instead of modifying the initial noise $\mathbf{z}_T$, we apply a gentle guidance force during the denoising process, steering the generation towards a final latent $\mathbf{z}_0$ whose DCT coefficients match our target signal $\mathbf{S}$. This guidance is applied for steps $t > T \cdot r_{\text{start}}$, where r_{start} is a ratio controlling the start of watermarking (e.g., 0.5).

At each guided denoising step t , the process is as follows (summarized in Algorithm 1):

Predict Clean Latent: Using the current latent $\mathbf{z}_t$ and the U-Net's noise prediction $\mathcal{E}_\theta$, we first estimate the predicted clean latent $\hat{\mathbf{z}}_0$ using Equation 1.

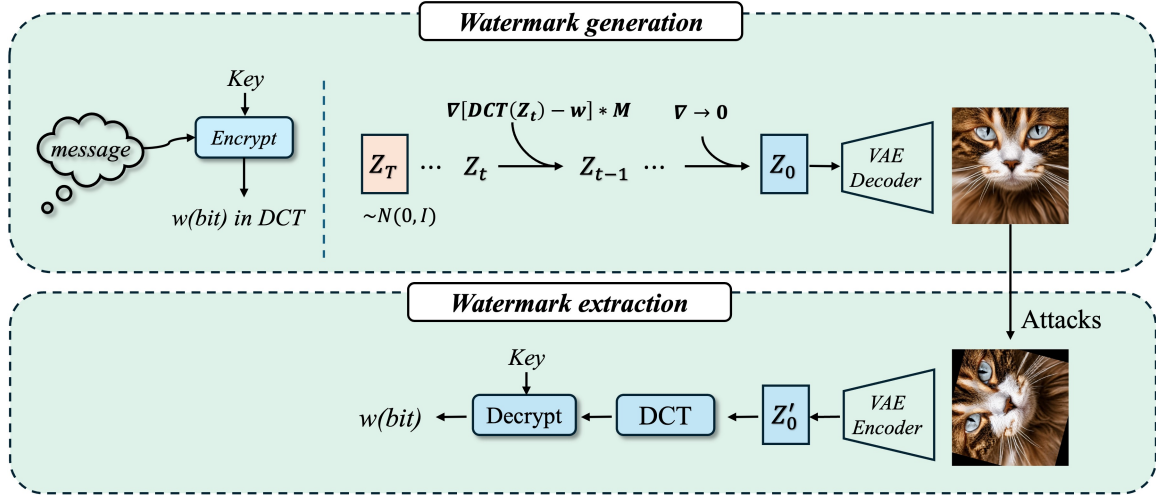


Figure 2. The overall framework of our proposed method, GROW. **(a) Watermark generation:** A secret message W is first converted into a bit sequence w . The watermark w is embedded into the target frequency domain at the positions defined by a mask M , which is generated from a secret key K . During the iterative denoising process, we compute a loss between the DCT of the predicted clean latent z_t and the target signal. The gradient of this loss guides the generation, progressively embedding the watermark. **(b) Watermark extraction:** To extract the watermark, the final image is encoded back to its latent representation z_0 . We then apply DCT and, using the key K , check the signs of the coefficients at the locations specified by the mask M . To ensure robustness, each bit of the message is embedded multiple times; a majority vote among these redundant copies reveals the final bit, making the extraction highly resilient to distortions. This process is direct and requires no inversion.

Calculate and Apply Gradient Guidance: We transform the predicted latent $\hat{z}_0$ into the frequency domain using DCT. A watermark loss $\mathcal{L}_{wm}$ is computed as the Mean Squared Error (MSE) between the DCT coefficients of $\hat{z}_0$ and our target signal S , exclusively at the locations specified by the mask M :

$$\mathcal{L}_{wm} = \|(DCT(\hat{z}_0) - S) \odot M\|_2^2 \quad (5)$$

The gradient of this loss with respect to $\hat{z}_0$ is then computed and used to steer the prediction in a single step:

$$\hat{z}_0^{\text{guided}} = \hat{z}_0 - \eta \nabla_{\hat{z}_0} \mathcal{L}_{wm} \quad (6)$$

where η is the guidance scale.

Update Noise Prediction for Scheduler: The standard scheduler requires a noise prediction to compute the next latent z_{t-1} . Since we have modified the predicted clean latent to $\hat{z}_0^{\text{guided}}$, we must derive a new, corresponding noise prediction. By rearranging Equation 1, we compute this guided noise $\mathcal{E}^{\text{guided}}$.

$$\mathcal{E}^{\text{guided}} = \frac{1}{\sqrt{1 - \bar{\alpha}_t}} (z_t - \sqrt{\bar{\alpha}_t} \hat{z}_0^{\text{guided}}) \quad (7)$$

This new noise, which now implicitly contains the watermark signal, is then used as the conditional prediction $\mathcal{E}_\theta(z_t, t, c)$. The guided prediction is then used within the standard Classifier-Free Guidance (CFG) framework in Eq. 3 to compute the final noise prediction for the current step.

4.2.3. Watermark Extraction

The extraction process, summarized in Algorithm 1, requires no inversion. Given a potentially watermarked image x , we first obtain its latent representation z_0 using the VAE encoder. We then apply the DCT to z_0 to get its frequency representation F .

Concurrently, the secret key K is used to deterministically regenerate the binary mask M , which identifies the exact frequency locations where the watermark bits were embedded. For each location (u, v) where the mask $M(u, v)$ is 1, we extract a single bit by checking the sign of the corresponding DCT coefficient $F(u, v)$: a positive sign decodes to '1', and a negative sign decodes to '0'. Since each bit was repeated multiple times, we use a majority vote to determine the final value for each bit, enhancing the robustness of the extraction. Finally, the collected sequence of bits is converted back into the message W .

4.3. Analysis: Progressive Guidance is Effective

Progressive guidance succeeds where a naive one-step injection into the final latent z_0 fails catastrophically in Figure 6. The answer lies in how GROW harmonizes with the iterative nature of the diffusion process, which can be attributed to two key principles:

Incremental Perturbations: The naive approach applies a single, large perturbation, pushing the final latent point far off the learned data manifold. In contrast, GROW

Algorithm 1 GROW: Generation and Extraction

```
1: procedure GENERATE(prompt  $c$ , message  $W$ , key  $K$ , ...)
2:    $\mathbf{S} \leftarrow \text{GenerateTargetSignal}(W, K)$   $\triangleright$  Create target watermark signal
3:    $\mathbf{z}_T \sim \mathcal{N}(0, \mathbf{I})$ 
4:   for  $t = T, \dots, 1$  do
5:      $\mathcal{E}_\theta \leftarrow \text{U-Net}(\mathbf{z}_t, t, c)$ 
6:      $\hat{\mathbf{z}}_0 \leftarrow \text{PredictCleanLatent}(\mathbf{z}_t, \mathcal{E}_\theta, t)$ 
7:     if  $t > T \cdot r_{start}$  then  $\triangleright$  If in guidance phase
8:        $\mathbf{g} \leftarrow \nabla_{\hat{\mathbf{z}}_0} \text{Loss}(\text{DCT}(\hat{\mathbf{z}}_0), \mathbf{S})$   $\triangleright$  Compute watermark gradient
9:        $\mathcal{E}_\theta \leftarrow \text{ApplyGuidance}(\mathcal{E}_\theta, \mathbf{g}, \eta)$   $\triangleright$  Update noise with guidance
10:    end if
11:     $\mathcal{E}^{\text{final}} \leftarrow \text{ApplyCFG}(\mathcal{E}_\theta, \dots)$ 
12:     $\mathbf{z}_{t-1} \leftarrow \text{SchedulerStep}(\mathcal{E}^{\text{final}}, t, \mathbf{z}_t)$ 
13:  end for
14:  return  $\mathbf{x} \leftarrow \text{VAE.decode}(\mathbf{z}_0)$ 
15: end procedure

16: procedure EXTRACT(image  $\mathbf{x}$ , key  $K$ )
17:   $\mathbf{M} \leftarrow \text{GenerateMask}(K)$   $\triangleright$  Regenerate watermark mask
18:   $\mathbf{z}_0 \leftarrow \text{VAE.encode}(\mathbf{x})$ 
19:   $\mathbf{F} \leftarrow \text{DCT}(\mathbf{z}_0)$ 
20:   $\text{bits} \leftarrow \text{Sign}(\mathbf{F}[\mathbf{M}])$   $\triangleright$  Extract bits from signs at masked locations
21:   $W \leftarrow \text{BitsToMessage}(\text{bits})$ 
22:  return  $W$ 
23: end procedure
```

applies a series of subtle, gradient-based updates at each step. These small adjustments gently steer the generation trajectory along the manifold’s tangent space, preserving semantic integrity throughout the process.

Collaborative Refinement: Crucially, the U-Net does not treat these small perturbations as noise to be eliminated. Instead, leveraging its powerful learned priors, it interprets them as high-frequency details to be rendered. The network collaboratively weaves the watermark’s frequency signal into plausible image textures, such as the grain of wood or ripples in water rather than leaving them as artifacts.

5. Experiments

5.1. Datasets and Metrics

Datasets. To ensure a comprehensive evaluation, we conduct tests on 1,000 generated images for each method from two widely-used datasets: MS-COCO [26] for photorealistic images and Stable-Diffusion-Prompts [16] for artistic and complex concepts.

Metrics. We evaluate performance from two perspec-

tives: *Imperceptibility*. We measure the visual quality of watermarked images using PSNR, SSIM, Fréchet Inception Distance (FID) [17], and LPIPS [42]. For PSNR, SSIM, higher is better. For FID, LPIPS, lower is better.

Robustness to attacks. We evaluate the robustness of the embedded watermark by measuring the Message Accuracy (M-ACC) after applying various attacks. M-ACC is a strict metric that calculates the percentage of images from which the entire embedded message is recovered without any bit errors. This is significantly more challenging than Bit Accuracy and more meaningful for practical applications where message integrity is paramount. The attacks include Rotation, JPEG compression, Cropping, Gaussian Blur, Gaussian Noise, Brightness adjustment and adversarial attack Diff-Pure [30] (illustrated in Figure 3).

5.2. Implementation Details

All experiments are performed using the Stable Diffusion v2.1-base [33] model. We compare GROW against a wide range of watermarking methods, including post-processing based (DwtDctSvd [35], Hidden [44]) and other training-free methods (Tree-Ring [38], RingID [9], Gaussian Shading [40], WIND [2]).

All experiments are conducted using the PyTorch 1.13.0 framework, running on a single Tesla T4 GPU. For our method, we set the default parameters to $\alpha = 0.5$, $\eta = 100$, sample 50 steps and $r_{start} = 0.5$ to test. And our experiments embed a 16-bit watermark in the first channel of the VAE features. However, our framework is flexible, and other channels can also be utilized to embed watermarks, either independently or concurrently, allowing for higher-capacity watermarking.

5.3. Experimental Results

Imperceptibility. A critical requirement for any watermarking technique is that it must not degrade the visual quality of the generated content. As shown in Table 1,

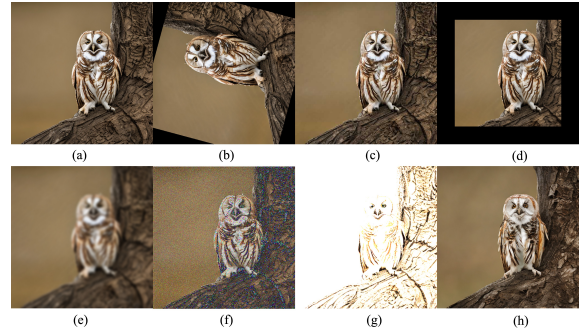


Figure 3. Watermarked image is attacked by different noise, including (a) with watermark, (b) Rotation 75°, (c) JPEG 25%, (d) Random Crop 25%, (e) GauBlur r=4, (f) Noise $\sigma=0.05$, (g) Brightness factor=6 and (h) Adversarial Attack $s=0.5$.

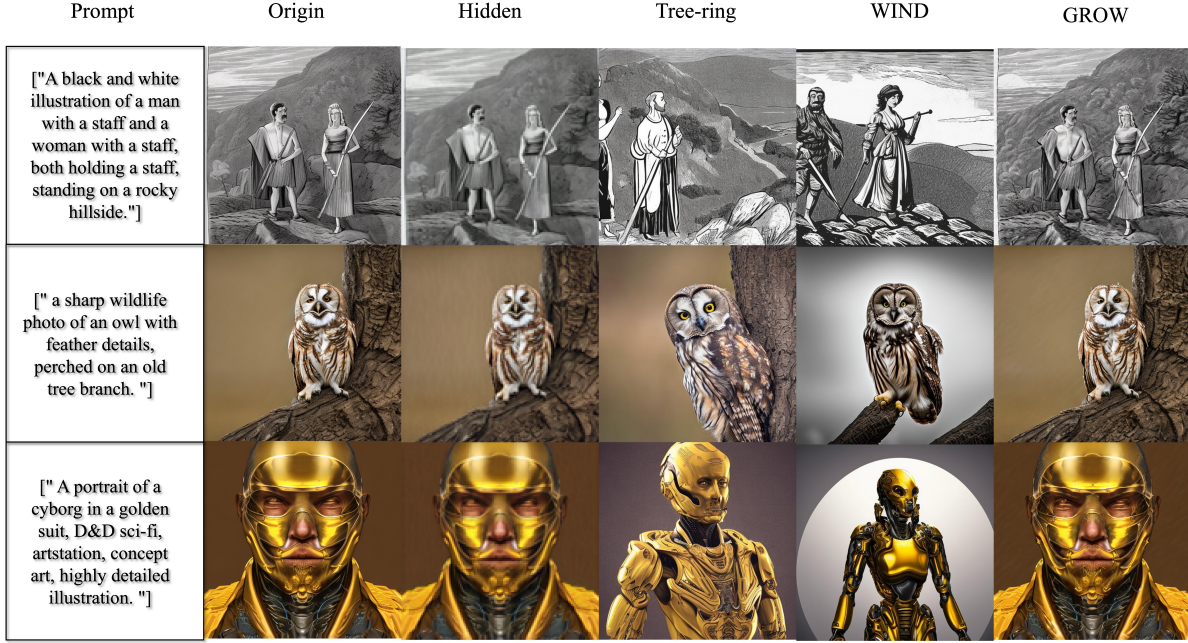


Figure 4. Visual comparisons of different methods using various prompts, including Hidden, Tree-Ring, WIND, and our GROW.

Table 1. Comparison of different watermarking methods across datasets. For imperceptibility metrics (PSNR, SSIM, FID, LPIPS), we report the performance on clean images. For robustness, we report the M-ACC (%) under various attacks. Best results are in **bold**, second best are underlined.

Method	Imperceptibility				Robustness to Attacks								
	PSNR $\uparrow$	SSIM $\uparrow$	FID $\downarrow$	LPIPS $\downarrow$	Clean	Rotation	JPEG	Crop	Blur	Noise	Brightness	Adversarial	Avg
<i>Dataset: Stable-Diffusion-Prompts</i>													
DwtDetSvd [35]	39.38	0.98	25.01	0.03	1.00	0.45	0.03	0.52	0.47	0.70	0.11	0.59	0.483
Hidden [44]	23.05	0.82	24.51	0.08	0.91	0.79	0.53	0.91	0.76	0.75	0.74	0.65	0.755
Tree-Ring [38]	15.25	0.54	25.93	0.42	0.99	0.02	0.42	0.04	0.61	0.53	0.42	0.85	0.485
RingID [9]	13.56	0.45	26.30	0.45	1.00	<u>1.00</u>	1.00	0.53	0.99	1.00	0.96	0.87	0.918
Gaussian Shading [40]	24.83	0.81	25.45	0.07	1.00	0.58	1.00	<u>0.92</u>	0.91	0.96	0.98	<u>0.91</u>	0.907
WIND [2]	13.16	0.47	<u>24.12</u>	0.39	<u>1.00</u>	0.72	<u>1.00</u>	0.86	<u>0.99</u>	<u>0.98</u>	0.97	0.89	<u>0.926</u>
GROW	<u>28.09</u>	<u>0.84</u>	18.90	<u>0.06</u>	1.00	1.00	1.00	1.00	1.00	0.93	<u>0.94</u>	0.94	0.976
<i>Dataset: MS-COCO</i>													
DwtDetSvd [35]	42.05	0.99	24.65	<u>0.04</u>	1.00	0.47	0.06	0.50	0.45	0.72	0.14	0.56	0.487
Hidden [44]	23.75	0.80	23.21	0.09	0.94	0.76	0.57	<u>0.93</u>	0.75	0.79	0.71	0.67	0.765
Tree-Ring [38]	16.48	0.27	22.38	0.43	0.99	0.05	0.46	0.08	0.64	0.50	0.44	0.87	0.504
RingID [9]	15.73	0.51	23.63	0.48	1.00	<u>1.00</u>	1.00	0.57	0.97	0.98	<u>0.97</u>	0.88	0.921
Gaussian Shading [40]	26.21	0.83	22.45	0.07	1.00	0.59	1.00	0.92	0.94	0.97	0.96	<u>0.93</u>	0.913
WIND [2]	7.44	0.26	<u>21.89</u>	0.44	<u>1.00</u>	0.75	<u>1.00</u>	0.84	<u>0.98</u>	<u>0.97</u>	0.96	0.90	<u>0.925</u>
GROW	<u>27.54</u>	<u>0.85</u>	12.32	0.05	1.00	1.00	1.00	1.00	0.99	0.91	0.98	0.95	0.978

GROW demonstrates excellent performance in preserving image fidelity. High PSNR and SSIM scores, along with a low LPIPS value, indicate that our method does not introduce significant pixel-level distortions and that the watermarked images are structurally very similar to the originals. Furthermore, GROW achieves a remarkable FID score of **12.32** on MS-COCO, signifying that the distribution of our watermarked images closely matches that of authentic, non-watermarked images, thus preserving the high genera-

tion quality of the underlying diffusion model. The qualitative comparisons in Figure 4 visually confirm the superior imperceptibility of our method. **Robustness.** As presented in Table 1, GROW demonstrates superior performance across the board. On the Stable-Diffusion-Prompts dataset, GROW achieves an average robustness M-ACC of 97.6%, and on MS-COCO, it reaches a higher 97.8% against a wide array of common distortions. This surpasses other methods, particularly recent training-free approaches.

Table 2. Comparison of different watermarking methods across diffusion models. For imperceptibility (PSNR, SSIM, FID, LPIPS), we report performance on clean images. For robustness, we report the average Message Accuracy (M-ACC) across all attacks. Best results are in **bold**, second best are underlined.

Method	Imperceptibility				Robustness to Attacks								
	PSNR $\uparrow$	SSIM $\uparrow$	FID $\downarrow$	LPIPS $\downarrow$	Clean	Rotation	JPEG	Crop	Blur	Noise	Brightness	Adversarial	Avg
<i>Model: stable-diffusion-v1-5</i>													
Tree-Ring[38]	14.25	0.35	25.18	0.45	0.97	0.02	0.41	0.05	0.58	0.45	0.39	0.83	0.463
RingID[9]	13.85	0.44	26.45	0.49	1.00	<u>1.00</u>	1.00	0.51	0.94	0.99	<u>0.95</u>	0.85	0.898
Gaussian Shading[40]	<u>24.75</u>	<u>0.79</u>	24.85	<u>0.09</u>	1.00	0.52	1.00	<u>0.87</u>	0.89	0.94	0.93	<u>0.89</u>	0.883
WIND[2]	6.85	0.22	<u>23.15</u>	0.46	<u>1.00</u>	0.68	<u>1.00</u>	<u>0.79</u>	<u>0.96</u>	<u>0.95</u>	0.92	<u>0.86</u>	<u>0.901</u>
GROW	26.85	0.82	15.75	0.07	1.00	1.00	1.00	1.00	0.97	0.87	0.96	0.91	0.965
<i>Model: stable-diffusion-xl-base-1.0</i>													
Tree-Ring [38]	17.15	0.29	23.85	0.44	0.98	0.04	0.43	0.07	0.61	0.47	0.42	0.84	0.483
RingID[9]	16.25	0.53	24.75	0.47	1.00	<u>1.00</u>	1.00	0.55	0.95	0.98	<u>0.96</u>	0.86	0.913
Gaussian Shading[40]	<u>27.35</u>	<u>0.84</u>	23.15	<u>0.06</u>	1.00	0.57	1.00	<u>0.90</u>	0.92	0.96	0.95	<u>0.92</u>	0.915
WIND [2]	8.15	0.28	<u>22.45</u>	0.43	<u>1.00</u>	0.73	<u>1.00</u>	0.82	<u>0.97</u>	<u>0.96</u>	0.95	0.89	0.928
GROW	28.25	0.86	13.45	0.04	1.00	1.00	1.00	1.00	0.98	0.90	0.97	0.94	0.981

Table 3. Time efficiency comparison (format: total generation time/watermark overhead and extraction time). All times are measured in seconds per image on a single Tesla T4 GPU. Our method avoids costly inversion, enabling fast watermark extraction.

Method	Generation Time (s) $\downarrow$	Extraction Time (s) $\downarrow$
Tree-Ring	21.03/ 0.04	19.70
RingID	21.07/0.08	15.30
Gaussian Shading	21.24/0.25	12.50
WIND	21.17/0.18	21.70
GROW (Ours)	21.16/0.17	0.24

Additionally, we conduct verification experiments in supplementary material to distinguish between watermarked and non-watermarked images.

Extraction Efficiency. Many contemporary methods, such as Tree-Ring, rely on the computationally expensive DDIM inversion process for extraction, making them unsuitable for real-time applications. All methods generate a watermark-free image under the same conditions takes 20.99 seconds. As shown in Table 3, all compared methods exhibit similar total generation times, as the minor overhead introduced by each watermarking algorithm is negligible compared to the 50-step DDIM sampling process. For instance, GROW’s progressive guidance adds only 0.17 seconds. The crucial advantage lies in extraction: GROW’s inversion-free mechanism is exceptionally efficient, achieving an extraction time of just 0.24 seconds, which is nearly 100x faster than inversion-based techniques. This dramatic efficiency gain makes GROW a viable and practical solution for deployment in production environments where fast verification is essential.

5.4. Different Diffusion Models

To assess the generalizability of our approach, we evaluated GROW and other methods on both stable-diffusion-v1-5 [32] and the more advanced stable-diffusion-xl-base-1.0 [1] (SDXL) model. The comprehensive results in Table 2 confirm that GROW’s advantages are model-agnostic. Across both architectures, GROW consistently achieves a balance of high imperceptibility and robustness. This suggests that the progressive guidance mechanism is a fundamental and portable technique for watermarking diffusion models.

5.5. Hyperparameters Analysis

In each experiment, we vary a single parameter while keeping others at their default optimal values ($\alpha = 0.5, \eta = 100, r_{start} = 0.5$). The results, which plot the trade-off between robustness (Accuracy) and image quality (FID), are shown in Figure 5.

Effect of Watermark Strength (α). As illustrated in Figure 5(a), the watermark strength α directly controls the trade-off between robustness and imperceptibility. Increasing α from 0.3 to 0.7 boosts the average M-ACC from 92% to 99%, as a stronger signal is more resilient to attacks. However, this comes at the cost of image quality, with the FID score degrading from 9.5 to 28.2.

Effect of Guidance Scale (η). The guidance scale η determines how forcefully the watermark is injected at each step. Figure 5(b) shows that increasing η from 50 to 100 significantly improves M-ACC from 90% to 99%. Beyond this point, accuracy saturates. Interestingly, the FID score shows that image quality is optimal around $\eta = 125$ (FID 15.1) before slightly degrading, suggesting that an excessively high scale can be counterproductive.

Effect of Start Ratio (r_{start}). The start ratio r_{start} defines at which point in the diffusion process the watermarking guidance begins. As shown in Figure 5(c), applying the

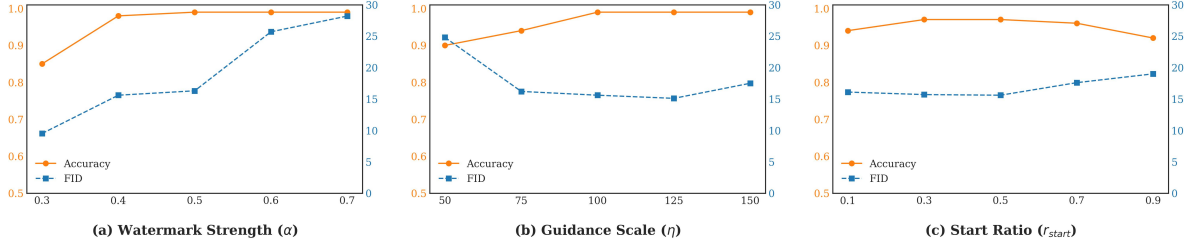


Figure 5. Ablation studies on key hyperparameters. We plot the trade-off between image quality (FID, blue line) and robustness (M-ACC, orange line). (a) Watermark strength α . (b) The guidance scale η . (c) Start Ratio.

Table 4. Quantitative comparison of image quality for the progressive guidance ablation study.

Method	FID ↓	PSNR ↑	SSIM ↑
One-Step	157.8	11.36	0.19
GROW	12.32	27.54	0.85

guidance earlier (i.e., a smaller r_{start}) enhances robustness. For example, starting at $r_{start} = 0.3$ yields a high M-ACC of 97%. This is because an earlier start allows the watermark to be integrated over more diffusion steps, weaving it more deeply into the image’s semantic structure. Conversely, starting too late (e.g., $r_{start} = 0.9$) results in a noticeable drop in robustness (92%).

5.6. Ablation Study on Progressive Guidance

To validate the necessity of our progressive guidance mechanism, we conduct a crucial ablation study against a simplified ”one-step” baseline. This baseline injects the same DCT-based watermark pattern directly into the final latent space (z_0) just before decoding, using an identical watermark strength. The results are striking. As shown in the visual comparison in Figure 6, the one-step method completely corrupts the image content. We quantify this degradation on MS-COCO dataset using standard image quality metrics in Table 4. The one-step method leads to a catastrophic collapse in image quality, with FID soaring while PSNR and SSIM plummet. This experiment unequivocally demonstrates that progressively integrating the watermark is essential for embedding a strong signal without a disastrous loss of image quality.

6. Limitations and Future Work

Despite its strong performance, GROW, as a training-free method in the DCT domain, has limitations. Its robustness can be compromised by random ratio stretching, and its capacity is inherently constrained by the frequency domain structure. A promising future direction is to replace the static DCT domain with a learnable, attack-resilient repre-

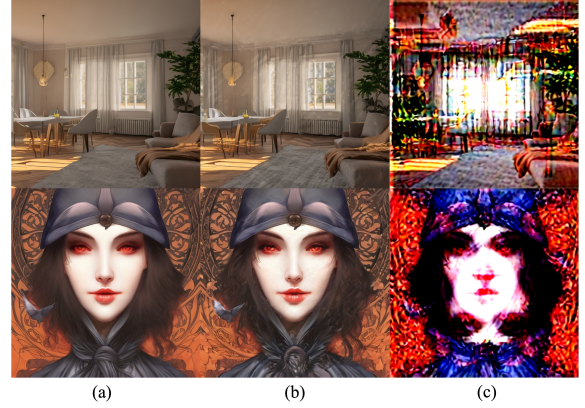


Figure 6. A visual comparison using prompts from the MS-COCO dataset. (a) Non-watermarked image. (b) Watermarked image by GROW. (c) One-step method.

sentation space. By training a dedicated encoder, one could generate a robust watermark embedding space optimized against a wide range of attacks. This learned space could then be integrated into our progressive guidance framework, potentially creating a new generation of watermarks with significantly enhanced security and robustness.

7. Conclusion

We proposed GROW, a novel and effective training-free watermarking paradigm for diffusion models. By reformulating watermark embedding as a progressive guidance task, our method iteratively guides the generation to form a watermark throughout the denoising process. This multi-step integration strategy ensures that the watermark is deeply fused with the generated content, achieving an excellent balance between imperceptibility and robustness. Extensive experiments validate that GROW significantly surpasses existing methods, demonstrating resilience to a wide range of attacks while preserving the high fidelity of the original diffusion model. Furthermore, by bypassing the computationally expensive inversion step required by many prior methods, GROW enables highly efficient watermark extraction.

Acknowledgements

This work was completed at the Pattern Recognition Center, WeChat AI, Tencent. The authors would like to express their sincere gratitude to Professor Anil K. Jain for his valuable guidance, insightful discussions, and continuous encouragement, which greatly benefited this work.

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